MILITARY SPECIFICATION
FOR
FLASK, AIR, LAUNCHER

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - This specification defines the requirements for manufacture, est, and shipment of curved, cylindrical flasks used in submarines for storing compressed air or gas at 4,500 pounds-per-square-inch pressure.

2. APPLICABLE DOCUMENTS

2.1 Documents - The following documents, to the extent specified herein, of the issue in effect on the date of the contract, form a part of this specification.

SPECIFICATIONS

MILITARY

MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for

NAVY DEPARTMENT'

BUREAU OF SHIPS	General Specification for Building Ships of the United States Navy, Section S9-1, Welding and Allied Processes

FSC 1440

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STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

NAVY DEPARTMENT

BuWep 2023561	Air Flask, Launch - Ejector Group Assembly
Buwep 2024826	Air Flask - Launch
BuWep 2024898	Air Flask, Launch - Ejector Group Assembly
BuWep 2024899	Air Flask - Launch

PUBLICATIONS

NAVY DEPARTMENT

BUREAU OF NAVAL WEAPONS

OP 400	General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment
NAVORD INST 4355.28	Equipment for Use by Government Inspectors in Final Inspection of Naval Ordnance Items
BUWEPS INST 4355.5	Bureau of Naval Weapons Calibration Program

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General Specifications for Inspection & Material, and Appendix II - Metals, Part A, Definitions and Tests

Standards Laboratory Information Manual

(Copies of specifications, standards drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Unfired Pressure Vessel Code

(Application for copies should be made to American Society of Mechanical Engineers, Boiler and Pressure 'Vessel Code Committee, 29 West 39th Street, New York 18, New York.)

AMERICAN IRON AND STEEL INSTITUTE

Specification 4130

(Application for copies should be made to American Iron and steel Institute, 150 East 42nd Street, New York 17, New York.)

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AMERICAN SOCIETY FOR TESTING MATERIALS

Specification A-372, Class 4	Carbon and Alloy Steel Forgings for Pressure Vessel Shells

(Application for copies should be made to American Society for Testing Materials, 1916 Race Street, Philadelphia, Pennsylvania.)

NATIONAL BUREAU OF Standards

Handbook H28	Screw-Thread Standards for Federal Services

(Application for copies should be made to Superintendent of Documents, Government Printing office, Washington 25, D.C.)

3. REQUIREMENTS

3.1 Description - The air flask is seamless-steel pipe having a capacity of 28.5 cubic feet $\pm$ 2 percent and shaped to conform to the contour of the submarine hull. The air flask is used for storing compressed air or gas at a working pressure of 4,500 psi, at temperatures up to +200 degrees Fahrenheit.

3.2 General Requirements

3.2.1 Preproduction Samples - Unless otherwise specified in the contract or order, preproduction samples of the air flask shall be manufactured using the methods and procedures proposed for the production lot. These samples will be tested as specified in section 4 herein and are for the purpose of determining that the production item meets the requirements of this specification (refer to 4.3.2). At the discretion of the procuring agency, the requirement for a preproduction sample may be waived and shall then be satisfied by substitution of a production unit chosen at random by the Government Inspector.

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3.2.2 Materials - Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the air flask shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in BuWep Dwg 2023561 and 2024826 or 2024898 and 2024899.

3.2.3 Finishes, Coatings, Construction, and Dimensions - Finishes, coatings, construction, and dimensions for the air flask shall be in accordance with the applicable drawings, specifications, and publications listed in BuWep Dwg 2023561 and 2024826 or 2024898 and 2024899.

3.2.4 Marking - Marking of the air flask shall be done by stamping in accordance with BuWep Dwg 2023561 and 20248260F 2024898 and 2024899.

3.2.5 Interchangeability - Individual piece parts of the air flasks fabricated in agreement with the applicable drawings and specifications, shall be interchangeable with like numbered parts.

3.2.6 Gages - The contractor shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings. If so stipulated in the contract, requisition, or order, the Government will furnish drawings of pertinent Navy Final Inspection Gages for guidance in the design of the contractor's inspection gages. However, such procedure shall not serve to relieve the contractor of’ his responsibility in the design and manufacture of such gages as may be required for the satisfactory fulfillment or the contract requirements but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.

3.2.7 Threads - Unless otherwise specified, all threads shall be in accordance with the National Bureau of standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

3.2.8 Request for Deviations - The procedures for requesting deviations from this specification or any specification, publication, material, or process specified herein shall be as set forth in OP 400.

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3.2.9 Conflicting Requirements - In event of conflict between this specification and any specification, publication, or drawing referenced herein, this specification shall govern.

3.2.10 Tolerances - Unless otherwise specifically noted, tolerances and accuracies stated herein are absolute.

3.3 Performance Requirements

3.3.1 Hydrostatic Pressure - The air flask shall withstand a hydrostatic pressure of 7,500 psig for 5 minutes, shall not rupture, and shall show no permanent set or change in external or internal volume exceeding 10 percent of the resulting expansion. If permanent set within the 10-percent limit is noted after testing, reapplication of the 7,500 psig pressure shall produce no additional permanent set.

3.3.2 Leakage - The air flask shall withstand air or gas at 4,500 psig, shall show no leakage around the neck and belled sections, and shall be tested as specified in paragraph 4.5.4.2.

3.3.3 Surface Quality - The air flask shall be free of cracks and voids and shall be tested as specified in paragraph 4.5.4.3. ,

3.3.4 Bending Quality - The steel used in air-flask fabrication shall not fracture or show other defects when tested as specified in paragraph 4.6.2.

3.3.5 Tensile Strength - Unless-otherwise specified in the contract, the flask shall have a yield point of not less than 80,000 psi, an ultimate tensile strength of not less than 105,000 psi, an elongation of not less than 15 percent in 2 inches, and shall be tested by the method specified in paragraph 4.6,3.

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3.4 Environmental Conditions

3.4.1 Corrosive Atmosphere - The exterior surfaces of the air flask shall withstand sea-atmosphere conditions, shall show no corrosion, and thereafter shall be capable of meeting all performance requirements of this specification.

3.4.2 Corrosion Resistance, Flask Interior - The interior surface of the air flask shall withstand contact with sea water and entrapped ambient atmosphere for 30 days, shall show no corrosion, and thereafter shall be capable of meeting all performance requirements of this specification.

3.5 Cleanup - prior to and after final assembly, all parts components, and the assembly shall be thoroughly cleaned of foreign matter. Burrs and sharp edges shall be removed.

3.6 Workmanship - The equipment, including all parts and accessories shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be paid to marking of parts and assemblies weldings and freedom of parts from burrs and sharp edges. Welding shall comply with Section S9-1 of General Specifications for Building Ships of the United States Navy.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's Inspection - Unless otherwise specified herein, the contractor is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

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4.2 Government Inspection - The Government Inspector will make such inspections as are necessary to determine that components and assemblies are in accordance with the requirements of the contract, pertinent drawings, and specifications. Unless otherwise specified, these inspections will be conducted in accordance with instructions contained in OP 400. Production tests of final assemblies shall be on a 100-percent basis. No sampling is permitted for acceptance inspection purposes. Only those assemblies which pass final inspection shall be accepted by the Navy; a rejected assembly may be reworked and resubmitted in accordance with OP 400. Failed parts and components shall be submitted to the Material Review Board for disposition.

4.2.1 Preproduction Sample - The preproduction sample specified in 3..l shall be submitted to the Government Inspector for preproduction tests at an activity designated in the contract or by the Bureau of Naval Weapons for purposes of determining compliance with the requirements of the contract, specifications, and drawings. Further production of the item by the contractor prior to approval of the contracting activity, or the completion of tests on the preproduction sample, shall be at the contractors risk. Preproduction samples accepted will be applied as part of the quantity specified by the contract or order.

4.3 Classification of Tests - The inspection and testing of the air flask shall be classified as follows:

4.3.1 Production Tests - These tests are to be accomplished on each air flask being submitted for acceptance under contract. Production tests shall be either performed by the manufacturer and witnessed by the Government Inspector, or these tests may be conducted by the Government Inspector. These tests are detailed in 4.5.

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4.2 Preproduction Tests - Preproduction tests are those which are accomplished on samples selected as specified in 4.2.1 which are representative of the production of the air flask to determine that the production meets the requirements of this specification. These are detailed in 4.5 and 4.6. Acceptance shall be based on no defects in the sample. Failure of the sample to comply with these requirements will result in the cessation of production until determination of cause of failure is made and appropriate correction approved by the procuring activity.

4.3.3 Tests After Delivery - Unless otherwise specified in the contract, the Government reserves the right to conduct any or all of the tests specified herein as deemed necessary after installation of the air flask. Any defects discovered before, during, or after these tests, due to the contractor's failure to comply with drawings or specifications for this equipment shall be corrected by the contractor at no expense to the Government, provided the contractor is notified of such defects within the guarantee period of the contract.

4.4 Special Test Equipment - No special test equipment is required to perform the acceptance tests set forth in this specification.

4.5 Production Tests

4.5.1 Responsibility - The contractor or Government Inspector shall conduct production tests as specified. in paragraph 4.3.1 to assure that the air flask is in compliance with the requirements of 3.3.1 through 3.3.3.

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4.5.2 Test Conditions - Unless otherwise specified, the air-flask shall be subjected to production tests under the following conditions:

Temperature:	Room ambient, +65 to +85 degrees Fahrenheit
Altitude:	Normal, ground atmospheric pressure
Vibration:	None
Humidity:	Relative humidity of 50 percent or less at +78 degrees Fahrenheit

4.5.3 Test and Inspection Equipment and Facilities - The manufacturer shall furnish and maintain all necessary test equipment, facilites, and personnel for performing all production tests. The test equipment shall be adequate in quantity and, unless definite requirernents are specified, shall be of sufficient accuracy and quality to permit performance of the required production tests.

4.5,3.1 Calibration - Measuring equipment and instrumentation shall be calibrated as required by NAVORD INST 4355.1, BUWEPS INST 4355.5, and in accordance with the policies specified in the Standards Laboratory Information Manual, issued by the Bureau of Naval Weapons Representative, Pomona, California.

4.5.4 Test Procedures

4.5.4.1 Hydrostatic-Pressure Test - To meet the requirements of 3.3.1, each air flask shall be subjected to a hydrostatic pressure of 7,500 psig for 5 minutes, shall not rupture, and shall meet the dimensional requirements of BuWep Dwg 2023561 and 2024826 or 2024898 and 2024899. Any permanent set within the specified tolerance shall be cause for reapplication of the 7,500 psig test pressure for 5 minutes. Any permanent set exceeding the amount specified in 3.3.1, due to the reapplication of the test pressure, shall be cause for rejection.

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4.5.4.2 Leakage Test - To meet the requirements of paragraph 3.3.2, each air flask shall be subjected to an internal air pressure of 4400-4600 psig to test for leakage around the neck and belled section. Any evidence of leakage shall be cause for rejection.

4.5.4.3 Magnetic Particle Test - Each air flask, before being coated, shall be magnetic particle inspected in accordance with Standard MIL-STD-271.

4.6 Preproduction Tests

4.6.1 Chemical Analysis Test - To meet the requirements of 3.2.2. drillings shall be taken out of pipe from each heat of steel and submitted to a Government laboratory for chemical analysis. The pipe shall be stamped by the Government Inspector for identification.

4.6,2 Bend Test - To-meet the requirements of 3,3,4, a transverse test specimen, of suitable length and from 1 to 2 inches in width, shall be cut from the wall of the sample flask. The corners of' the cross section of the specimen may be broken with a smooth file, but no appreciable rounding of the corners will be permitted. The specimen shall be bent inwardly until the greatest dimension between the outside surface does not exceed six times the thickness of the specimen, The specimern shall not fracture or show other defects.

4.6.3 Tension Test - Four tension-test specimens shall be cut longitudinally from the sample flask selected by the Government Inspector, at points approximately 90 degrees from one another. The tension tests shall be conducted in accordance with Appendix II, Part A of General Specifications for Inspection of Materials; the specimens shall be Type 4 as specified therein and shall meet the requirements of 3.3.5.

4.6.4 Corrosive Atmosphere Test - The air flask shall be subjected to a simulated sea atmosphere in accordance with Specification MIL-E-5272 to meet the requirements of 3.4.1.

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4.6.5 Corrosion Resistance Test - To meet the requirements of 3.4.2, one air flask, mounted in the normal position and partially filled with sea water, shall be allowed to stand for 30 days and shall show no corrosion.

4.7 Rejection - Failure of the air flask to meet any of the requirements and tests of this specification shall be considered cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging - Preservation and packaging of the air flask shall be Level A or Level C, as follows:

5.1.1 Level A - Level A shall be as specified by the procuring activity.

5.1.2 Level C - Cleaning, drying, preservation, and unit packaging shall be in accordance with the manufacturer's commercial practice. For domestic shipment and immediate use, the neck opening of the air flask shall be sealed with a blank flange, fitted with a spring-loaded air check valve, and charged with dry, nitrogen gas to a pressure of 7 psig. Any preservative applied shall be to exterior surfaces only.

5.2 Packing - Packing of the air flask shall be Level A or Level C, as follows:

5.2.1 Level A - Level A shall be as specified by the procuring activity.

5.2.2 Level C - The air flask, packaged as specified in 5.1.1, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classification Rules or other common-carrier regulations applicable to the mode of transportation.

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5.3 Marking - In addition to any special markings required in the contract or order, marking of the air flask and shipping containers shall be in accordance with Standard MIL-STD-129. Type, weight, capacity (in cubic feet), stock number, and length shall be included in the unit markings. Flask markings shall be on waterproof tags secured to the neck of the flask with corrosion-resistant wire. Nomenclature shall be the exact nomenclature which has been approved under the contract or order.

6. NOTES

6.1 Ordering Data - Procurement documents should specify the specific title, number, and date of this specification and exceptions to this specification, applicable drawings, and other documents.

6.2 Material - The material called for on BuWep Dwg 2024826 is as follows:

(a) Flask shell material is seamless--steel pipe in accordance with ASTM Specification A-372 Class 4, except minimum yield point shall be 80,000 psi in the normalized condition.

(b) Flange material is alloy steel in accordance with AISI Specification 4130, heat treated to 80,000 psi minimum yield point.

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Shell and Head Thickness - The thickness of material shown on BuWep Dwg 2024826 and 2024899 was determined as follows:

(a) Shell Thickness - Shell thickness was determined in accordance with the ASME Unfired Pressure Vessel Code using a corrosion allowance of 1/16 inch. The maximum allow- stress used in calculating the shell thickness shall provide a minimum factor of safety of 2.62 based on the ultimate tensile strength of the material and a minimum factor of 2 based on the minimum yield strength of the material.

(b) Head Thickness - Outlet head thickness shall be the same as the shell thickness. The opposite head shall have a minimum thickness as specified by the ASME Unfired Pressure Vessel Code with a l/16-inch corrosion allowance.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procure-merit operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings , specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation. or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto

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